

**TAMPINES JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008**

**BIOLOGY
Paper 3 Applications Paper**

9747/03

Question 1

After nearly a decade of setbacks and false starts, stem-cell science finally seems to be hitting its stride. Just a year after Japanese scientists first reported that they had generated stem cells by reprogramming adult skin cells — without using embryos — American researchers have managed to use that groundbreaking technique to achieve another scientific milestone. They created the first nerve cells from reprogrammed stem cells — an important demonstration of the potential power of stem-cell-based treatments to cure disease.

Led by Kevin Eggan at the Harvard Stem Cell Institute and Christopher Henderson at Columbia University, the 13-person team reported online today in *Science Express* that they had generated motor neurons from the skin cells of two elderly patients with a rare form of ALS, or Lou Gehrig's disease, a progressive neurodegenerative condition. The new study marks an important first step on the road toward real stem-cell-based therapies, and also answers several plaguing questions about the pioneering stem-cell technique known as induced pluripotent stem cell, or iPS, generation.

IPS was first described by Japanese biologist Shinya Yamanaka, who, in 2007, showed that the introduction of four genes into an adult human skin cell could reprogram it back to an embryonic state. Like embryonic stem cells, these reprogrammed adult cells could be coaxed into becoming any other type of cell — from skin to nerve to muscle.

Adapted from article from time.com

- (a)(i) Embryonic stem cells can be isolated for research. State the location from which they can be isolated.

- **Inner cell mass from The blastocyst**

[1]

- (ii) Describe the unique properties of stem cells that make them useful in research.

- **Self renewal → able to be maintained in culture**
- **Can be pluripotent/multipotent**
- **Unspecialized cells that are able to be induced to differentiate into specialized cells**

[2]

Any 2 (1 m each)

- (iii) The passage mentioned 'induced pluripotent stem cells' (IPS)/reprogrammed stem cells that can be induced to form nerve cells to treat neurodegenerative diseases. Describe how this can be done *in vitro*.

- Induced stem cells are placed in culture media that contains nutrients and growth factors;
- to maintain the stem cell culture/allow proliferation of stem cells. ;
- Specific growth factors and hormones will be added to the culture media to induce differentiation of IPS → nerve cells ;
- Differentiation involved the turning on certain genes to give nerve cells their specific characteristic/function ;

[3]

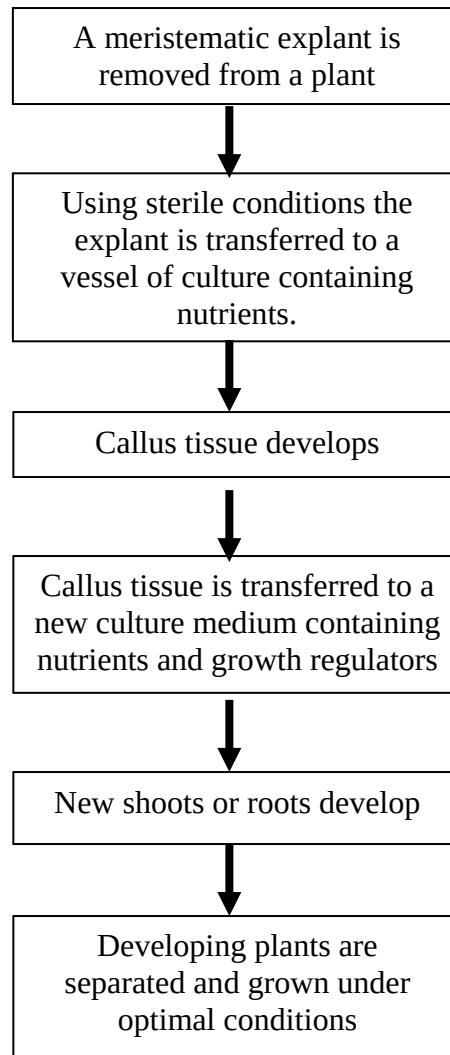
Any 3 of 4 points (1 m each)

-
- (iv) During reprogramming in adult cells to form reprogrammed stem cells, some genes must be turned on. Suggest a gene that must be turned on during this process.

- Telomerase gene
- ½ m if telomerase given only

[1]

- (b) Plant tissue culture is a method used to propagate plants. The flow diagram shows one method of plant tissue culture.



- (i) What is a callus?

- mass of undifferentiated cells ;
- that divide by mitosis ;
- totipotent ;that can be induced to
- differentiate ; into roots and shoot by varying the concentration of plant regulators/hormones in the culture media

[2]

$\frac{1}{2}$ m each

- (ii) What is a 'meristematic explant' and give one advantage in using an explant from the meristematic region for tissue culture?

- meristematic explant is explant /parts of plant taken from region with actively dividing cells ; like root tips / shoot tips ; (1 m)
- Advantage is the ability to produce disease-free plants (1 m)

[2]

- (iii) Tissue culture is often used to propagate many genetically identical plants from the parent plant.
Describe briefly the process of this propagation with the use of callus tissue that has been isolated from the parent plant.

- **Poliferation of callus tissue by subculturing ; to increase the number of calli available**
- **Transfer the callus to culture media ; with nutrients an varying concentration of plant growth hormones to induce callus differentiation**
- **High auxin:cytokinin → root formation ;**
- **High cytokinin:auxin → shoot formation ;**

$\frac{1}{2}$ m each

- (iv) Another type of plant cell culture is the culture of protoplasts from plants.
If I want to combine the good traits of two related plant species together, protoplast fusion would be a more suitable method.
Outline briefly the process of protoplast fusion.

- **Protoplasts are plant cells without cell walls;**
- **Extracted plant parts e.g leafs will be treated with enzymes like cellulose ; to digest the cell walls.**
- **Protoplasts from two plants can be induced to fuse ;**
- **With the addition of chemicals to form a hybrid protoplast ;**
- **This hybrid protoplast will contain the genetic materials from both donor plants.**

$\frac{1}{2}$ m each

Total: 15 marks

Question 2

Severe Combined Immunodeficiency disease (SCID) is a genetic disease that involves no production of immuno-competent of T and B lymphocytes, hence compromising the immune system of the patient. Opportunistic infections can kill the SCID patients before the first birthday.

Two types of SCID namely X-linked SCID and ADA SCID are the most common form of SCID.

Figure 2.1 shows the inheritance of the disease in a family in which both parents are carriers.

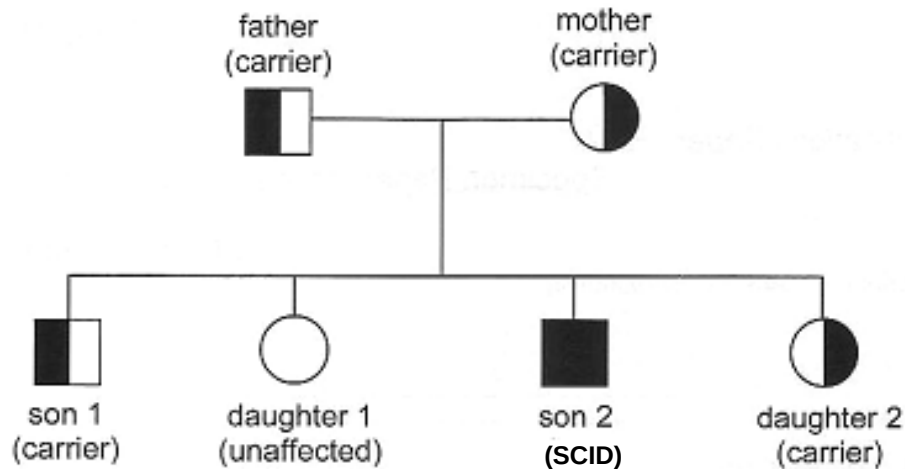


Fig. 2.1

(a)(i) State the form of SCID that is being inherited in the family.

- **ADA deficiency SCID**

[1]

(ii) Explain your answer in (a)(i) using the diagram in Fig. 2.1.

- **The pedigree shows a recessive inheritance pattern.**
- **Son 2 has inherited 2 recessive alleles from each parent.**
- **Not sex linked because son 1 is a carrier, not affected by the disease.**
- **If sex linked, son 1 will inherit the recessive allele from the mother → will have SCID.**

[2]

$\frac{1}{2}$ m each

(iii) Describe briefly the genetic basis of X-linked SCID.

- **Mutation in the interleukin 2 receptor gamma chain gene found on the X chromosome ;**
- **defective gamma chain → defective interleukin 2 receptor;**
- **Ligands like interleukins cannot bind → no transduction pathway initiated;**
- **No differentiation and production of mature T cells from immature T cells;**
- **Immune system will be defective.**

[4]

1 m each

2 RFLP morphs have been found tightly linked to the ADA gene on chromosome 20 which can be used for genetic screening. A radioactive probe has been designed to reveal the RFLP #1 and RFLP #2.

The arrows on Fig. 2.1 show the position of the restriction sites of EcoRI. The asterisk (*) indicate the position of a mutation at the restriction site that results in the loss of the restriction site.

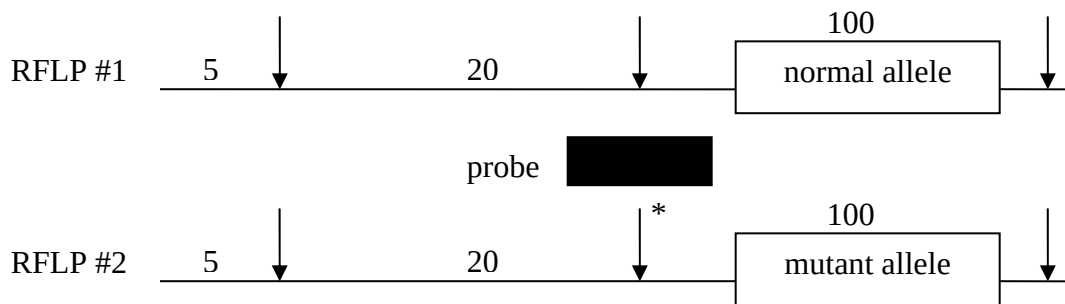


Fig. 2.1

In the space provided below, draw the resulting autoradiogram you would expect to see.

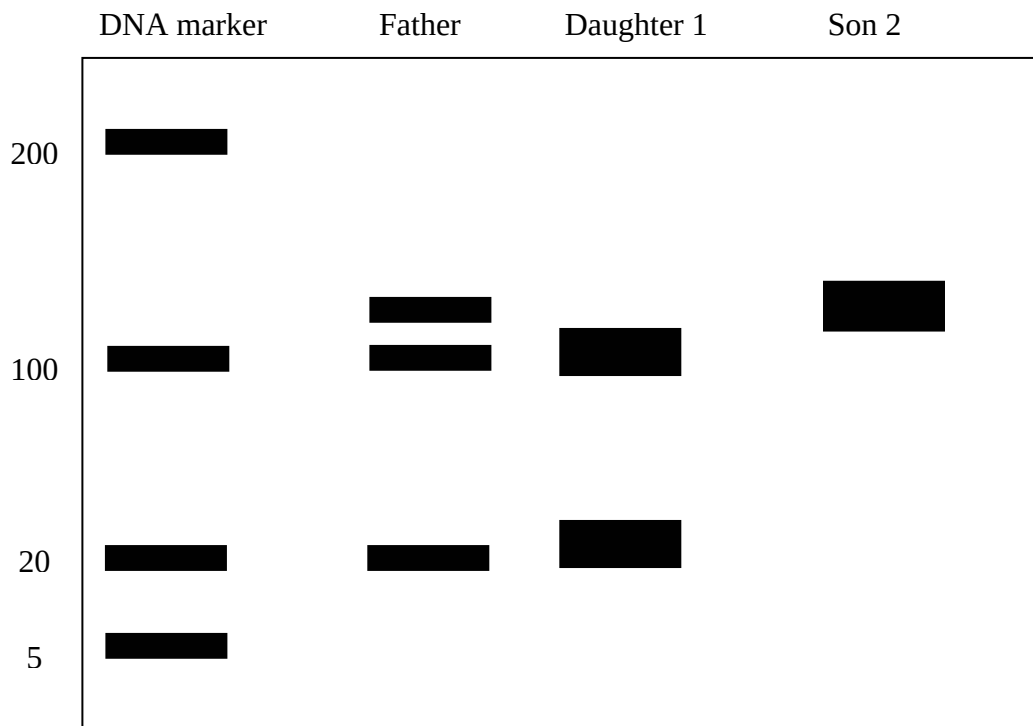


Fig.2.2 (1 mark for daughter, 1 mark for son)

Mark for correct position, no penalization for thickness of bands

[2]

(ii) Explain how a probe hybridizes to a DNA fragment.

- Probe is complementary to the sequence of the DNA fragment
- Hybridized by complementary base pairing through the formation of hydrogen bonds.

[2]

(iii) Explain how DNA fragments can be separated into its bands.

- DNA fragments are separated according to size and charge in gel electrophoresis ; [2]
- DNA fragments are negative and will move to the positive electrode when an electric field is applied ;
- Larger fragments will move slower → higher band ;
- Smaller fragments will move faster → lower band ;

½ m each

(iv) Explain how differential banding patterns can result when chromosome 20 is isolated from the father, daughter 1, son 2 and cleaved with the same restriction enzyme, *EcoRI*? OR

- Digestion of chromosome 20 with the normal ADA allele will result in 2 bands of 100kb and 20 kb after autoradiography [2]
- On the chromosome with the mutant ADA allele → mutation has resulted in the loss of restriction site → only 1 band of 120kb will be visualized after autoradiography.
- Since father is heterozygous, daughter is homozygous dominant and son is homozygous recessive
- Digestion with *EcoRI* will result in the different position and thickness of the bands after autoradiography.

½ m each

Total: 15 marks

Question 3

Rice is the staple diet in many parts of the world. It lacks a number of important nutrients, including the β carotene, from night blindness. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

Figure 3.1 shows an artificial DNA sequence used.

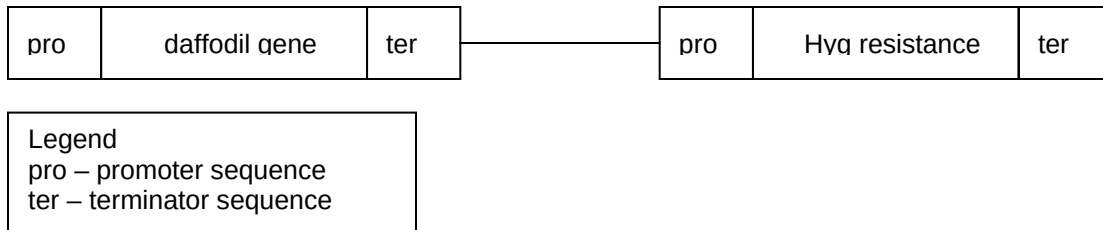
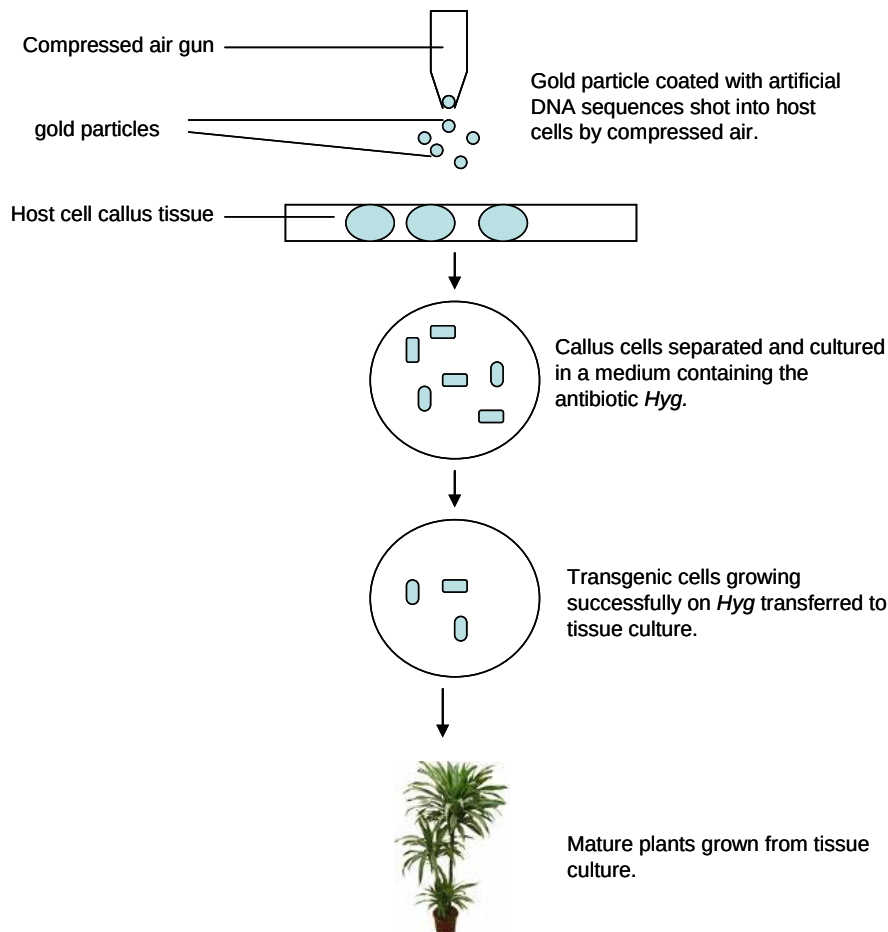


Fig 3.2 shows the main events in obtaining a transgenic plant.



- (a) With reference to figures 3.1 and 3.2,
- (i) Outline how the daffodil gene might have been isolated from the donor organisms and inserted into the artificial sequence.
- Daffodil gene is isolated with the help of specific restriction enzymes which cleave at specific sites by the breaking of phosphodiester bonds ; [2]
 - Southern blotting used ;
 - Include gel electrophoresis to separate the fragments according to sizes and blotting to a nitrocellulose membrane ;
 - Use of a specific probe to identify the fragment with daffodil gene ;

$\frac{1}{2}$ mark each

- (ii) Explain what is meant by :

Transgenic plant:

- foreign gene from another organism/species inserted into host cell/plant ; [1]
- foreign will be expressed to give a functional protein product/production of the foreign protein ;

$\frac{1}{2}$ m each

- (iii) Explain the role of *Hyg* resistance gene in this procedure.

- *Hyg* resistance gene serve as a selectable marker here. [2]
- Used in the selection process to select for recombinant cells
- Cells which have taken in the recombinant plasmid will produce the resistant protein to *Hyg* antibiotic
- And can survive on medium containing the *Hyg* antibiotic and be propagated.

$\frac{1}{2}$ m each

- (iv) Another method of producing a transgenic plant is by the use of bacterium named *Agrobacterium*.

Using any other named example, outline the process of using *Agrobacterium* to produce a genetically modified crop to improve the yield/nutrition of a plant.

- Any named example, insect-resistant (Bt) / herbicide resistant crop (cotton, corn, soybean, tomato plant) [4]

Marking answer:

- Gene of interest (β -carotene/Bt) and Ti plasmid ; cut with same restriction enzymes to create fragments with sticky ends ;
- Recombinant plasmid ; formed by ligation with DNA ligase ;
- Transformation ; of *Agrobacterium* with recombinant plasmid and selection ; for bacterium with recombinant plasmid
- Infection of plant cells (callus/protoplast) ; with recombinant *agrobacterium* and tissue culture ; to produce a whole plant.

1/2 m each

Process of genetic engineering in detail:

- The gene for Bt toxin (from *Bacillus thuringiensis*) / the gene encoding glyphosate resistant gene is isolated.
 - The gene of interest is joined to a tumour inducing / Ti plasmid that has been cut with the same restriction enzyme using DNA ligase
 - The resultant recombinant plasmid is transformed into *Agrobacterium tumefaciens*
 - Selection for *Agrobacterium* cells that have successfully taken up the recombinant plasmid, which are then cultured with leaf discs/callus tissue/protoplast.
 - Recombinant plant cells are then transferred to culture medium and propagated using tissue culture techniques.
-

- (b) Tomato plants were transformed by adding an antisense gene to pTOM13 to the normal genome. Antisense genes, once introduced, are inherited. Ethene production during fruit ripening was measured in normal plants and plants with one antisense gene. The results are shown in Fig. 3.3

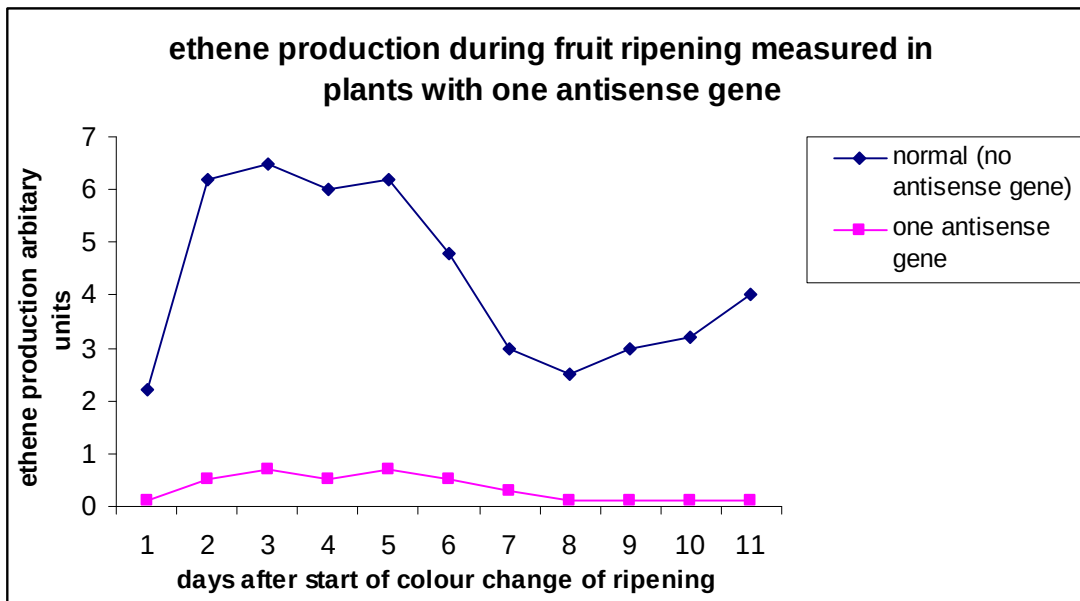


Fig. 3.3

- (i) With reference to Fig 3.3,
Comment/describe the effect on ethene production in the transgenic plant with the antisense gene.

Observations: (any two with values: 2marks)

- Production of ethene is higher in the normal plant as compared to the transgenic plant throughout the 11 days. (give difference → 1m)
- Lowest ethene production in normal plant at about 2 units at day 1 and highest at 6.2 a.u at day 3 as compared to the ethene production in the transgenic plant which stays fairly constant at about 0.2 a.u (lowest) to about only 1 a.u (highest)
- High fluctuations in the ethene production in normal plant but only slight fluctuations in the transgenic plant. (with shape → 1m)
- Steep increase in the ethene production from 2 a.u (day 1) to 6.5 a.u (day 3) and then gradual decrease to about 2.5 a.u (day 8) before an increase again to 4 a.u at day 11 for the normal plant. Ethene production is constant at about 0.2 a.u for the transgenic plant.

Conclusion(1 mark)

- Presence of antisense gene in the transgenic plant decreases the ethene production during the ripening process
- by keeping the expression of the protein responsible for ripening (pTOM13) low all the time.

[3]

(ii) Suggest a function of product of pTOM13 gene in the normal plant.

- Enzyme ; involved in the pathway of ethene production ;
 - Substrate for enzyme to make ethene
 - (pdt ;) that (control ethene production ;)
-

[1]

(iii) Describe briefly how the introduction of an antisense gene into the tomato plant reduces the production of ethene in the ripening process.

- Antisense gene is transcribed to mRNA ;which is complementary to
- the mRNA transcribed from the pTOM13 gene by RNA polymerase.
- Complementary base pairing ; between the antisense mRNA and the
- pTOM13 mRNA to form a duplex structure.
- pTOM12 mRNA is prevented from binding to the ribosome ; → no
- translation
- No pTOM13 protein product / no translation of the pTOM13 mRNA →
- reduction of production of ethene.

[2]

$\frac{1}{2}$ m each

Total: 15 marks

Essay question

Question 1

- (a) Describe features of a plasmid that made it suitable as a vector. You may want to use a diagram for illustration. [4]
- (b) Using a named example, outline the process of cloning a eukaryotic gene into *E.coli* cells to produce eukaryotic protein in genetic engineering. [10]
- (c) Discuss the social and ethical implications of production of genetically modified organisms to solve the demand for food in the world. [6]

Proposed answer scheme for essay question

(a) Describe features of a plasmid that made it suitable as a vector. You may want to use a diagram for illustration. [4]

1) Origin of replication;

- to allow the plasmid to replicate in concert with the host DNA;
- GOI replicated along with vector

2) Selectable / Genetic Marker (2 markers involved)

- Usually an antibiotic resistant gene e.g. ampicillin resistance / Lac Z gene;
- 1st marker to select for bacterial cells which have taken in the plasmid / successfully transformed bacterial cells
- 2nd marker to select for recombinant bacterial cells / cells that have taken in the recombinant plasmid.
- Via replica plating / blue-white colonies selection ;

3) Multiple cloning/restriction sites

- found within the second selectable marker (e.g. tetracycline or Lac Z gene)
- allows insertion of GOI that treated with different R.E
- Insertion will disrupt the function of the selectable marker e.g tetracycline resistance gene or Lac Z gene that allows selection for recombinant plasmid.

(1/2 mark for each point)

(1 mark for labeled drawing)

(b) Using a named example, outline the process of cloning a eukaryotic gene into bacteria cells to produce eukaryotic protein in genetic engineering for therapeutic purposes. [10]

Example: (1 m for name)

- **Insulin** treat diabetes patients by maintaining blood glucose levels.
- **Human growth hormone** to treat dwarfism
- **Other valid example**

1. Construction of recombinant DNA molecule (foreign gene + vector DNA)

a. Source of gene

mRNA for insulin (1/2 m each pt, 2 max)

- mRNA **from** pancreas/ beta cells of pancreatic cells;
- **RT** is used to reverse transcribe mRNA to form complementary DNA (cDNA);
- **Alkaline/RNase H** to degrade the mRNA strand;
- **DNA pol** then synthesizes the second DNA strand;
- Mentioned use of linkers to create sticky ends (since cDNA will be blunt);
- Example of linkers;

b. Creation of recombinant DNA molecule using recombinant DNA technology

(1/2 m each pt, 2 max)

- Vector and insulin gene must be cleaved with same restriction enzyme at specific recognition sequence/site;
- So that complementary base pairing can form via hydrogen bonds between insulin gene fragment and vector;
- Sticky ends increases efficiency;
- Reason for use of vectors (unstable linear insulin gene/ OriR/ selectable marker gene);
- Example (of vector/ restriction enzyme);
- DNA ligase is added to seal phosphodiester bond;

2. Introduction of recombinant DNA molecule into living host (e.g. bacteria) **(1/2 m each pt, 2 max)**

- Via transformation;
- Method of transformation;
- Reason of method (to create permeability/ transient enlargement of pores);
- Example of host cell;
- Purpose of living host (provide enzyme machinery);

3. Cell growth/ division to form a colony; **(1 m)**

4. Screening/ Selection & identification of clones with recombinant DNA molecules

(1/2 m each pt, 2 max)

- Insertional inactivation;
- Lac Z/ Blue-white selection or antibiotic selection;
- Description of either lac Z or antibiotic selection ;

5. Purification of insulin product

(1/2 m each pt, 1 max)

- Upscale/ production in biofermenters with appropriate nutrient both/medium (max 1 mark);
- Extraction and purification of insulin protein from bacteria cells to be packaged.
- Post-translational modification/ combining the 2 chains via disulphide bonds;

(c) Discuss the social and ethical implications of production of genetically modified organisms to solve the demand for food in the world. [6]

1) Definition of GMO – **1m**

Genetically Modified Organism: Organism that has acquired one or more genes by artificial means. The genes may or may not be from the same species.

2) Social implications - **(each point 1 m with elaboration up to a max of 3m)**

(a) Threat to Human Safety

(i) *Genetic markers*

- Vectors used for transforming plant/crop cells – contain antibiotic markers/ kanamycin resistance; (example is Flavr Savr tomatoes)
- Concern that the antibiotic resistant gene (if it is not properly broken down by the digestive system) may be passed to potentially harmful bacteria making them resistant to the antibiotics;

(ii) *Allergens or Toxic/carcinogenic*

- Transgenic food can be potential sources of allergens.
- Eg. those in peanuts can cause allergic reactions in some people.

(b) Threat to Safety of the Environment

(i) *Genetically modified crops might establish themselves as weeds*

- Seeds from the transgenic crops that have been modified to withstand unfavourable environmental conditions, may be carried by the wind to other places. These crops grow quickly and may established themselves as weeds.
- *Spread of resistance from genetically modified crops to weeds*
- Pollen grains from genetically modified plants (eg. Herbicide resistant plant) can be carried in the wind and fertilise wild relatives and weeds.
- This may result in “superweeds” –weeds that are resistant to herbicide. These may become more invasive (overrun agricultural areas rapidly) or compete with the natural species.

(iii) *Upsetting the ecological balance*

- the genetically modified organisms (eg. Giant salmons) themselves may be more aggressive thus causing fear that they may upset the balance of the native species.
- decrease in biodiversity;

(iv) Use of herbicide

- another concern is that developing herbicide-resistant plants may encourage the use of greater amounts of herbicides;
- the possibility of pests developing resistance to toxin produced by crop plants eg. Bt toxin;

3) Ethical implications (**each point 1 m with elaboration up to a max of 3m**)

(a) Is genetic manipulation of animals and plants in principle acceptable? Do we have the right to tamper with nature by mixing genes among species?

(b) Is it acceptable to engineer any organism to be used as food for human consumption.

- For example, the use of BST in dairy cattle increases the risk of mastitis. There seems to be *little concern about whether the animals are biologically 'designed'* to withstand the additional burden of increased milk or egg or meat production.

(c) *Religious concerns or dietary restrictions*

- Eg. Plants that have been modified to carry animal genes or vice versa;

(d) Should anyone be able to patent a genetically modified animal or plant?

- Many research companies have sought to patent the transgenic plants or animals that they have developed. Some people argued that patenting animals is itself unethical as it reduces them to the level of objects. If patenting is not allowed, how else could a company protect the results of their research program?

(e) Is genetic engineering to make a staple crop more resist in marginal conditions (e.g. drought, cold) a potential boon for Third World agriculture, or another danger of increased dependence on rich "developed" countries?

- i.e. richer countries benefiting at the expense of the poorer countries;
- world food production may be dominated by a small number of large companies with the technical know-how;

(f) How great are the potential risks involved in releasing genetically modified organisms into the biosphere without knowing all the possible consequences?

(g) Should products on sale be labeled to indicate that genetic engineering was involved in their production?

